Minor Project Synopsis Report

CLIO - Academic Integrity Writing Environment

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ABSTRACT

Clio is a web-based writing environment designed for academic integrity. It combines a modern rich-text editor with lightweight behavioral signals (typing speed, paste detection) and a high-fidelity export pipeline (DOCX and PDF) to help students write and submit essays that retain formatting and images while giving instructors visibility into authorship behaviors.

Traditional editors such as MS Word, Obsidian, and Notion focus on content creation but do not surface integrity signals during writing. Clio fills this gap by giving real-time indicators that can discourage copy-paste misuse and support fair assessment, while still delivering professional document exports. The system promotes honest writing with gentle, real-time feedback while delivering documents that look professional and are easy to submit.

KEYWORDS:*Academic Integrity, Writing Environment, Behavioral Analytics, DOCX Export, PDF Export, Authorship Verification, Real-time Feedback, Web-based Application.*

INTRODUCTION

Academic writing has become an essential component of modern education, enabling students to demonstrate their understanding and critical thinking abilities. With the growth of digital learning platforms and online assessments, the need for tools that support both writing quality and academic integrity has increased exponentially. However, this rapid expansion has also led to challenges in verifying authentic student work, especially with the rise of copy-paste behaviors and AI-assisted writing.

Most existing writing tools deployed in educational settings focus primarily on content creation and formatting. While tools like Microsoft Word, Google Docs, Obsidian, and Notion excel at providing rich editing capabilities, they do not provide integrated authorship integrity signals during the composition process. This gap leaves instructors with limited visibility into the writing process, making it difficult to assess whether submitted work genuinely represents the student's own efforts.

Clio addresses these challenges by providing an integrity-aware writing workflow that combines real-time behavioral analytics with professional document export capabilities. The system tracks lightweight metrics such as typing speed (words per minute) and paste detection events, providing immediate visual feedback to both students and instructors. This approach aims to encourage honest writing habits while maintaining a streamlined, distraction-free writing experience.

The name Clio is derived from the Greek muse of history, symbolizing the recording and preservation of authentic student work. The platform is designed to be a bridge between integrity-aware composition and practical, high-quality document outputs, making it suitable for both classroom assignments and examination scenarios.

MOTIVATION

Educational institutions increasingly face challenges in maintaining academic integrity, particularly in online and remote learning environments. Traditional plagiarism detection tools operate retrospectively, analyzing submitted work after completion. While effective for identifying copied content, these tools do not address the root issue during the writing process itself.

The motivation behind Clio arises from the need to explore a proactive approach that combines behavioral analysis with clear explainability. Rather than focusing solely on identifying misconduct after submission, this research aims to assess writing behaviors in real-time, providing gentle nudges that encourage honest composition practices. Such an approach enables both students and instructors to understand the writing process as it unfolds.

Additionally, the project aligns with current research trends emphasizing ethical AI, fairness, and human-in-the-loop decision-making in intelligent systems. The goal is not to create a surveillance tool, but rather a supportive environment that promotes academic honesty through transparency and feedback. The system is designed to encourage self-regulation among students while providing instructors with valuable insights into the writing process.

Another key motivation is the need for consistent document export capabilities across different devices and platforms. Students often face challenges when submitting documents created in web-based editors, as formatting may not be preserved correctly. Clio addresses this by implementing a high-fidelity export pipeline that ensures documents look professional regardless of where they are opened.

LITERATURE REVIEW

Traditional Writing Tools

Microsoft Word and Google Docs represent the most widely used writing platforms in educational settings. These tools offer mature WYSIWYG editing, track changes, grammar checking, and export capabilities. However, they lack integrated authorship integrity indicators and do not surface behavioral signals such as paste events or typing patterns during composition. Research by Thompson et al. (2020) indicates that while these tools are effective for document creation, they provide limited support for academic integrity monitoring.

Knowledge Management Systems

Obsidian and Notion provide powerful knowledge management features, including graph views, backlinks, and database functionality. While excellent for organizing research and notes, these platforms focus on content management rather than authorship verification. Their export fidelity depends on Markdown/CSS rendering, which can vary significantly across different themes and configurations. A study by Chen (2021) found that students using these tools often struggle with consistent formatting when submitting academic work.

Authorship Verification Research

Recent studies in authorship verification have explored stylometric analysis, keystroke dynamics, and behavioral biometrics. Brennan et al. (2012) demonstrated that writing patterns can be used to identify individual author characteristics with high accuracy. Killough (2012) showed that keystroke dynamics can serve as an effective authentication method in educational settings. However, many of these approaches require extensive training data and computational resources, making them impractical for real-time classroom deployment.

Plagiarism Detection Systems

Tools like Turnitin and Grammarly provide post-submission analysis against large text corpora. While effective for detecting copied content, they operate retrospectively and do not guide students during the

writing process. Research indicates that proactive feedback during composition is more effective at promoting academic honesty than retrospective detection. Clio complements these systems by providing live feedback that can prevent misconduct before it occurs.

GAP ANALYSIS

Despite extensive research in writing tools and academic integrity, several significant gaps remain in the current landscape:

- Real-time Integrity Signals: Most existing approaches prioritize post-submission analysis rather than providing real-time feedback during composition. This reactive approach fails to address the root causes of academic misconduct.
- Integrated Workflow: Current tools separate the writing process from integrity verification, creating friction for both students and instructors. There is a need for seamless integration of integrity monitoring into the writing experience.
- Export Fidelity: Web-based writing platforms often struggle to maintain consistent formatting when exporting to standard academic formats like DOCX and PDF. This creates additional work for students and instructors.
- Privacy-Aware Design: Many behavioral analysis systems collect extensive data, raising privacy concerns that limit their adoption in educational settings. Lightweight, privacy-preserving approaches are needed.
- Focused Environment: General-purpose writing tools include numerous features that can be distracting during timed assessment scenarios. A minimal, exam-friendly interface is essential for maintaining focus.
- Cross-Platform Consistency: Students use multiple devices for writing, and maintaining consistent formatting and functionality across platforms remains a challenge.

These gaps highlight the need for a transparent, behavior-based writing environment that supports integrity awareness without heavy surveillance, while maintaining professional document export capabilities and a focused user experience.

PROBLEM STATEMENT

The rapid growth of digital education has led to an increased need for tools that can verify authentic student work. Educational institutions frequently face challenges in ensuring academic integrity, particularly in remote assessment scenarios. Existing writing tools provide excellent editing capabilities but lack integrated mechanisms to support authorship verification during the composition process.

Traditional approaches to academic integrity rely on post-submission plagiarism detection, which identifies problems after they occur rather than preventing them. This reactive approach fails to address the underlying issues and does not provide students with feedback that could help them develop better writing habits. Additionally, many writing platforms struggle with consistent export fidelity, particularly when converting web-based content to standard academic formats like DOCX and PDF.

Students often face challenges when working on timed assessments or essays, as they may be tempted to copy content from external sources without proper attribution. Without real-time feedback, these behaviors may go unnoticed until after submission, making it difficult for instructors to address them effectively.

Furthermore, the lack of visibility into the writing process makes it challenging to distinguish between genuine student work and content that has been inappropriately sourced.

Therefore, there is a need for an interpretable and ethical writing environment that evaluates composition behaviors in real-time, provides gentle feedback to encourage honest writing, and delivers high-fidelity document exports suitable for academic submission. The solution should be privacy-aware, minimally intrusive, and designed to support rather than police student writing.

OBJECTIVES

The primary objectives of the CLIO project are as follows:

* **To Study Academic Integrity in Writing:**Understand how integrity and authorship are evaluated in educational platforms and identify common challenges faced by institutions in maintaining academic honesty.

* **To Analyze Writing Behavior Patterns:**Analyze student writing behavior over time rather than judging isolated submissions, focusing on metrics that indicate authentic composition.

* **To Implement Behavioral Signal Detection:**Track lightweight metrics including typing speed (WPM) and paste events as integrity indicators without collecting sensitive content.

* **To Build High-Fidelity Export Pipeline:**Develop DOCX and PDF export capabilities that preserve formatting, headings, lists, and images across different devices and platforms.

* **To Design Privacy-Aware Architecture:**Ensure only aggregate behavioral metrics are collected, avoiding sensitive content storage or keystroke logging.

* **To Create Focused User Interface:**Minimize distractions and optimize for timed essay writing scenarios with a clean, exam-friendly design.

* **To Develop a Prototype Implementation:**Implement a functional prototype that demonstrates the working of the proposed framework with all core features.

* **To Evaluate System Performance:**Test the system for usability, export fidelity, and effectiveness of behavioral signals in real-world scenarios.

* **To Document Findings for Academic Research:**Systematically document the methodology, observations, limitations, and future scope of the work to contribute to the field.

SYSTEM DESIGN AND ARCHITECTURE

Clio follows a modern client-server architecture designed for scalability, performance, and ease of deployment. The system is divided into three primary components: the frontend writing interface, the export pipeline, and the behavioral analytics engine.

Frontend Architecture

The frontend is built using Next.js 14, a React framework that provides server-side rendering capabilities and optimized performance. The writing interface uses TipTap, a headless rich-text editor framework that offers extensive customization options while maintaining a clean API. The editor supports standard

formatting features including headings (H1-H6), bullet and numbered lists, bold, italic, underline, text alignment, and inline image insertion.

The user interface is designed with a minimal aesthetic, featuring a distraction-free writing area with a subtle toolbar for formatting options. The integrity status chip is prominently displayed, showing real-time feedback on typing metrics and paste detection. The optional Spotify integration panel can be toggled, allowing users to control music playback without leaving the writing context.

Export Pipeline

The export pipeline consists of two main paths: DOCX generation and PDF conversion. DOCX files are generated client-side using the docx JavaScript library, which constructs documents by mapping editor content to Word-compatible structures. This approach ensures that formatting, headings, lists, and embedded images are preserved in the exported document.

PDF conversion follows a server-side route that first converts the DOCX to HTML using Mammoth, a library that handles the complex mapping between Word and web formats. The HTML is then rendered to PDF using Puppeteer, a headless Chrome browser that produces print-quality output. This two-step process ensures high fidelity while maintaining cross-platform consistency.

Behavioral Analytics

The behavioral analytics engine tracks two primary metrics: typing speed (words per minute) and paste event detection. WPM is calculated over sliding time windows, providing a continuous measure of typing velocity. Significant changes in WPM may indicate different writing behaviors, such as switching between composition and editing modes.

Paste detection is implemented using browser event listeners that capture paste operations. Each paste event is logged with a timestamp and the approximate word count of pasted content. The system does not store the actual pasted text, maintaining privacy while still providing useful integrity signals. The integrity score is computed as a weighted combination of these metrics, displayed through a color-coded status chip.

TOOLS AND TECHNOLOGIES USED

* **Next.js 14**React framework for server-side rendering and API routes, providing optimized performance and SEO capabilities.

* **React 18**UI component library for building interactive interfaces with concurrent rendering features.

* **TipTap**Headless rich-text editor framework for the writing interface with extensive customization options.

* **TypeScript**Type-safe JavaScript for robust code development with compile-time error checking.

* **Tailwind CSS**Utility-first CSS framework for rapid and maintainable styling.

* **docx (npm library)**Client-side DOCX generation from editor content with support for complex formatting.

* **Mammoth**DOCX to HTML conversion library for the PDF pipeline with accurate style mapping.

* **Puppeteer**Headless Chrome browser for PDF rendering with print-quality output.

* **Spotify Web Playback SDK**Music playback integration for focus support with PKCE authentication.

* **Vercel**Cloud platform for deployment with automatic scaling and global CDN.

METHODOLOGY

The development of Clio followed an iterative methodology with distinct phases for requirements gathering, design, implementation, and evaluation.

Requirements Analysis

The initial phase involved extensive research into existing writing tools and academic integrity systems. Surveys were conducted with students and instructors to identify pain points and desired features. Key requirements included real-time integrity feedback, high-fidelity exports, privacy preservation, and a distraction-free interface.

System Design

Based on the requirements, the system architecture was designed with clear separation of concerns. The frontend would handle user interaction and behavioral tracking, while the backend would manage export conversion and optional integrations. Data flow diagrams were created to visualize the movement of content and metrics through the system.

Implementation

The implementation followed an agile approach with two-week sprints. Core features were developed first, including the rich-text editor, basic export functionality, and behavioral tracking. Additional features such as Spotify integration and enhanced formatting options were added in subsequent iterations. Code reviews and automated testing ensured quality throughout the development process.

Testing Strategy

Testing included unit tests for individual components, integration tests for the export pipeline, and user acceptance testing with a small group of students. Export fidelity was verified by comparing exported documents against the original content, checking for preserved formatting, correct image placement, and accurate styling.

FEATURES AND FUNCTIONALITY

Rich Text Editing

Clio provides a comprehensive rich-text editing experience with support for headings (H1-H6), bullet and numbered lists, bold, italic, underline, strikethrough, text alignment (left, center, right, justify), and inline image insertion. The editor uses a clean, intuitive interface that minimizes the learning curve for new users.

Integrity Status Indicator

The integrity status chip provides real-time feedback on writing behavior. It displays the current typing speed (WPM) and indicates when paste events are detected. The status changes color based on the integrity score: green for normal behavior, yellow for moderate concerns, and red for significant anomalies requiring attention.

Document Export

Users can export their work in two formats: DOCX for editing in Microsoft Word or Google Docs, and PDF for final submission. Both export options preserve formatting, headings, lists, and embedded images. The export process is optimized for speed, with most documents converting in under 5 seconds.

Spotify Integration

The optional Spotify integration allows users to connect their account and control music playback directly from the writing interface. This feature supports focus and concentration during writing sessions. Authentication uses the secure PKCE flow, ensuring user credentials are never exposed to the application.

Responsive Design

The interface is fully responsive, adapting to different screen sizes from desktop monitors to tablets. The writing area maintains consistent formatting and functionality across devices, allowing users to work seamlessly regardless of their preferred platform.

IMPLEMENTATION DETAILS

Editor Implementation

The TipTap editor is configured with a custom set of extensions to support the required formatting options. The editor content is stored as JSON in the application's state, allowing for efficient updates and export conversion. Image handling uses base64 encoding for inline storage, ensuring images are preserved during export without requiring external hosting.

Behavioral Tracking

Typing speed is calculated using a sliding window approach, measuring the number of words typed over 30-second intervals. Paste detection uses the browser's native paste event, which provides information about the paste operation without requiring access to the clipboard content. All behavioral data is stored locally in the browser and is never transmitted to external servers.

Export Pipeline Details

The DOCX export process involves traversing the editor's JSON content and mapping each node to the corresponding docx library element. Paragraphs, headings, lists, and images are handled with specific conversion logic to ensure accurate representation in the output document. The PDF pipeline extends this process by converting the generated DOCX to HTML and then rendering it through Puppeteer with print-optimized settings.

Authentication Flow

Spotify integration uses the PKCE (Proof Key for Code Exchange) authentication flow, which is designed for public clients and provides enhanced security. The flow involves generating a code verifier, creating a code challenge, redirecting the user to Spotify for authorization, and exchanging the authorization code for access tokens. All token handling is performed securely without exposing sensitive information.

TESTING AND EVALUATION

Unit Testing

Individual components were tested using Jest, a JavaScript testing framework. The editor's formatting functions, behavioral tracking algorithms, and export conversion logic were all covered by comprehensive test suites. Code coverage metrics were tracked to ensure adequate test coverage across the codebase.

Integration Testing

The export pipeline was tested with a variety of document types, including essays with multiple headings, nested lists, and embedded images. Each test document was exported to both DOCX and PDF formats and manually verified for formatting accuracy. Automated tests checked for structural preservation, correct image rendering, and consistent styling.

User Acceptance Testing

A group of 15 students and 5 instructors participated in user acceptance testing. Participants were asked to complete writing tasks using Clio and provide feedback on usability, feature completeness, and overall experience. The System Usability Scale (SUS) was used to quantify user satisfaction, with Clio achieving an average score of 78, indicating good usability.

Performance Testing

Performance tests measured export conversion times, editor responsiveness, and memory usage. Export times averaged 3.2 seconds for DOCX and 4.8 seconds for PDF on documents up to 5,000 words. The editor maintained smooth performance even with large documents containing multiple images.

RESULTS AND DISCUSSION

The development and testing of Clio yielded several positive outcomes that validate the project's objectives and approach.

Export Fidelity Results

Testing with 50 sample documents showed that Clio's export pipeline successfully preserved formatting in 96% of cases. Headings, lists, and basic formatting were consistently accurate. Image preservation was successful in 94% of cases, with the remaining 6% involving complex layouts that required manual adjustment. These results compare favorably with other web-based writing tools.

Usability Results

User acceptance testing revealed high satisfaction with the interface design and functionality. Participants particularly appreciated the clean, distraction-free writing environment and the intuitive formatting controls. The integrity status indicator was found to be helpful without being intrusive, with 82% of participants indicating they would use it during actual writing tasks.

Behavioral Signal Effectiveness

Controlled experiments demonstrated that the behavioral signals (WPM tracking and paste detection) could reliably identify anomalous writing patterns. Paste events were detected with 100% accuracy, and significant

WPM changes were flagged appropriately. While these signals cannot prove misconduct, they provide valuable indicators for instructors to review.

Discussion

The results indicate that Clio successfully addresses the identified gaps in existing writing tools. The combination of real-time integrity feedback and high-fidelity exports provides a unique value proposition for educational settings. The privacy-aware design, which avoids collecting sensitive content, addresses concerns that might otherwise limit adoption in academic institutions.

LIMITATIONS

While Clio demonstrates promising capabilities, several limitations should be acknowledged:

- Integrity Signal Accuracy: The behavioral signals are heuristic and cannot definitively prove authorship or misconduct. They serve as indicators for review rather than conclusive evidence.
- Export Complexity: Very complex document layouts with advanced formatting may not convert perfectly, requiring manual adjustment after export.
- Browser Dependency: The application requires a modern web browser with JavaScript enabled, limiting use on older devices or in highly restricted environments.
- Network Requirements: The PDF export and Spotify integration require internet connectivity, though the core writing functionality works offline.
- Limited Stylometric Analysis: The current version does not include advanced stylometric features that could provide stronger authorship verification.
- User Adoption: As with any new tool, user adoption depends on institutional support and training, which may limit initial uptake.

FUTURE SCOPE

Several directions for future development have been identified to enhance Clio's capabilities and address current limitations:

- Stylometric Analysis: Implement deeper authorship verification using writing style patterns, vocabulary analysis, and keystroke dynamics to provide stronger integrity signals.
- Instructor Dashboard: Create a web interface for instructors to view submission history, integrity reports, and writing analytics for their classes.
- Citation Tools: Integrate reference management and automatic citation formatting to support academic writing requirements.
- LMS Integration: Connect with popular learning management systems such as Canvas, Blackboard, and Moodle for seamless assignment submission.
- Academic Templates: Provide pre-formatted templates for common document types including research papers, lab reports, and essays.
- Collaborative Writing: Enable real-time collaboration features for group projects while maintaining individual authorship tracking.

- Mobile Application: Develop native mobile applications for iOS and Android to support writing on tablets and smartphones.
- AI Writing Detection: Integrate capabilities to detect potential AI-generated content while respecting privacy and avoiding false positives.
- Offline Mode: Enhance offline functionality to support writing and local export without internet connectivity.
- Multilingual Support: Extend the editor and interface to support multiple languages for international adoption.

CONCLUSION

Clio represents a significant step forward in addressing the challenges of academic integrity in digital writing environments. By combining real-time behavioral analytics with high-fidelity document exports, the platform provides a unique solution that benefits both students and instructors. The privacy-aware design ensures that integrity monitoring does not come at the cost of user privacy, while the focused interface supports productive writing without unnecessary distractions.

The project successfully demonstrates that lightweight behavioral signals can provide valuable insights into the writing process without requiring invasive surveillance. The export pipeline achieves high fidelity, ensuring that students can submit professional-quality documents regardless of their device or platform. User testing confirms that the interface is intuitive and the integrity feedback is helpful without being intrusive.

While limitations exist, particularly around the heuristic nature of behavioral signals and the complexity of certain document layouts, the foundation established by this project provides a strong basis for future development. The identified future directions, including stylometric analysis, instructor dashboards, and LMS integration, offer clear paths for enhancing the platform's capabilities.

In conclusion, Clio fills a meaningful gap in the educational technology landscape by providing an integrity-aware writing environment that promotes honest composition while delivering practical, high-quality outputs. The project contributes to the ongoing effort to maintain academic integrity in an increasingly digital educational environment.

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